Tutorial 8

1. Let G be a simple graph. Show that the relation R on the set of vertices of G such that uRv if and only if there is an edge associated to {u,v} is a symmetric, irreflexive relation on G .

2. How many edges are there in a graph with 10 vertices each of degree six?

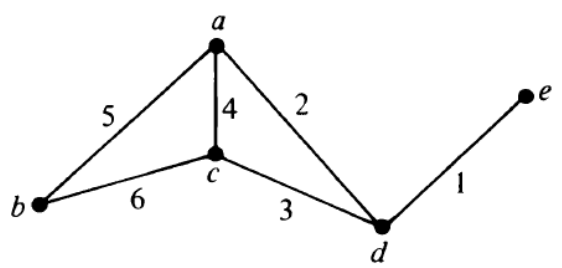
3. Which of the following statements is/are TRUE for an undirected graph?
   P: Number of odd degree vertices is even
   Q: Sum of degrees of all vertices is even
   (A) P Only
   (B) Q Only
   (C) Both P and Q
   (D) Neither P nor Q

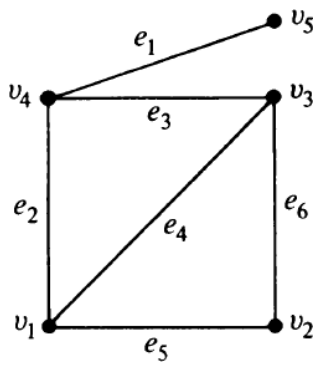
4. Prove that the sum of the degrees of the vertices of any finite graph is even.

5. Show that any graph where the degree of every vertex is even has an Eulerian cycle. Show that if there are exactly two vertices a and b of odd degree, there is an Eulerian path from a to b. Show that if there are more than two vertices of odd degree, it is impossible to construct an Eulerian path.

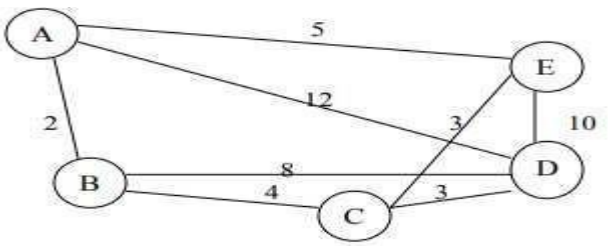
6. An n-cube is a cube in n dimensions. A cube in one dimension is a line segment; in two dimensions, it's a square, in three, a normal cube, and in general, to go to the next dimension, a copy of the cube is made and all corresponding vertices are connected. If we consider the cube to be composed of the vertices and edges only, show that every n-cube has a Hamiltonian circuit.

7. Show that following graphs are isomorphic:





8. The traveling salesman problem can be described as follows:
   TSP = {(G, f, t): G = (V, E) a complete graph, f is a function V×V → Z, t € Z, G is a graph that contains a traveling salesman tour with cost that does not exceed t}.
   Consider the following set of cities:



Find a minimal path passing from all vertices once for city A.

9. Draw an undirected graph represented by the given adjacency matrix.

$$\begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

10. Let G be a simple undirected graph. Let T_D be a depth first search tree of G. Let T_B be a breadth first search tree of G. Consider the following statements.

(I) No edge of G is a cross edge with respect to T_D. (A cross edge in G is between two nodes neither of which is an ancestor of the other in T_D).

(II) For every edge (u, v) of G, if u is at depth i and v is at depth j in T_B, then $|i - j| = 1$. Which of the statements above must necessarily be true?

(A) I only

(B) II only

(C) Both I and II

(D) Neither I nor II